Research Portfolio

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PROJECTS AND RESEARCHES

Studylytics Platform

Predictive Cognitive Profiling: Engineered a full-stack learning telemetry system mapping metrics to Cognitive Load Theory. Isolated an empirical -0.025 latency-accuracy paradox. Implemented Random

Edge AI Object Classifier

Win Research Centre: Fine-tuned a MobileNetV2 computer vision pipeline achieving 97.4% validation accuracy. Executed post-training integer quantization to achieve a 3.7MB binary footprint with an

Behavioral Analytics (ENP)

Longitudinal Case Study: Developed an integrated 10-column tracking schema evaluating multi-linguistic media consumption against qualitative journals over 6

Forest Classifiers yielding an 85% accuracy ceiling.

Review Technical Abstract →

ultra-low ~30ms inference latency threshold.

Review Technical Summary →

months (N=1). Empirically verified the Satiation Deficit Paradox.

Inspect Active Repository →

SELECTED PUBLICATIONS

Structural pipeline summaries and exploratory empirical commentaries are actively undergoing preparation for academic repository upload. Initial review articles are mirrored on Medium.

PROFESSIONAL VISION

Operating at the intersection of **Artificial Intelligence, Cognitive Science, and Human-Computer Interaction (HCI)**, I design full-stack data pipelines that quantify user cognitive load to engineer adaptive computational environments. My research methodology focuses on translating fine-grained behavioral interaction logs and longitudinal time-series data into actionable pedagogical and behavioral models. I am dedicated to constructing resource-efficient, human-centric technical architectures that improve digital learning paradigms and adaptive analytics.

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Mysuru, India • Integrated Research Hub v2.2

Studylytics: Cognitive Analytics

Pilot Research Study | Unique Attempts: n=135

This study investigates the intersection of Cognitive Science and Artificial Intelligence by modeling human learning behavior using interaction data. By tracking behavioral telemetry and contextual states, the platform quantifies cognitive load thresholds and performance variability in digital-native learners to overcome the academic supervision gap.

PREDICTIVE ACCURACY

85%

LATENCY CORRELATION

-0.025

REASONING ACCURACY

69%

ACADEMIC EVALUATION

99/100

KEY RESEARCH INSIGHTS

- **Cognitive Bottleneck:** Reasoning tasks required the longest processing duration (12.8s) yet produced the lowest accuracy parameters (69%), confirming abstract processes act as operational constraints.
- **The Latency-Accuracy Paradox:** A verified negative Pearson correlation (-0.025) demonstrates that prolonged response latency serves as a proxy for cognitive confusion rather than productive deliberation.

- **Mental Fatigue Threshold:** Efficiency peaks significantly at Focus Level 7 (93.3% accuracy) but sharply deteriorates at Focus Level 8 (86.7%), demonstrating diminishing performance returns due to system-driven over-analysis.
- **Feasibility of Predictive Profiling:** High precision (0.88) and recall (0.96) for correct assertions confirm that rolling metadata signals offer a reliable baseline to anticipate user performance.

METHODOLOGICAL BOUNDARIES & LIMITATIONS

The dataset is structurally restricted to a single-user profile, meaning conclusions serve as a baseline model rather than a generalized population sample. To systematically mitigate recollection bias (the Practice Effect), a manual stimulus rotation strategy was deployed every four sessions to confirm procedural learning. Predefined task labels showed calibration variances, indicating that future iterations require automated, data-driven difficulty balancing.

VISUAL PLOT DIAGNOSTICS

Figure 1: Performance Accuracy by Cognitive Task Variant

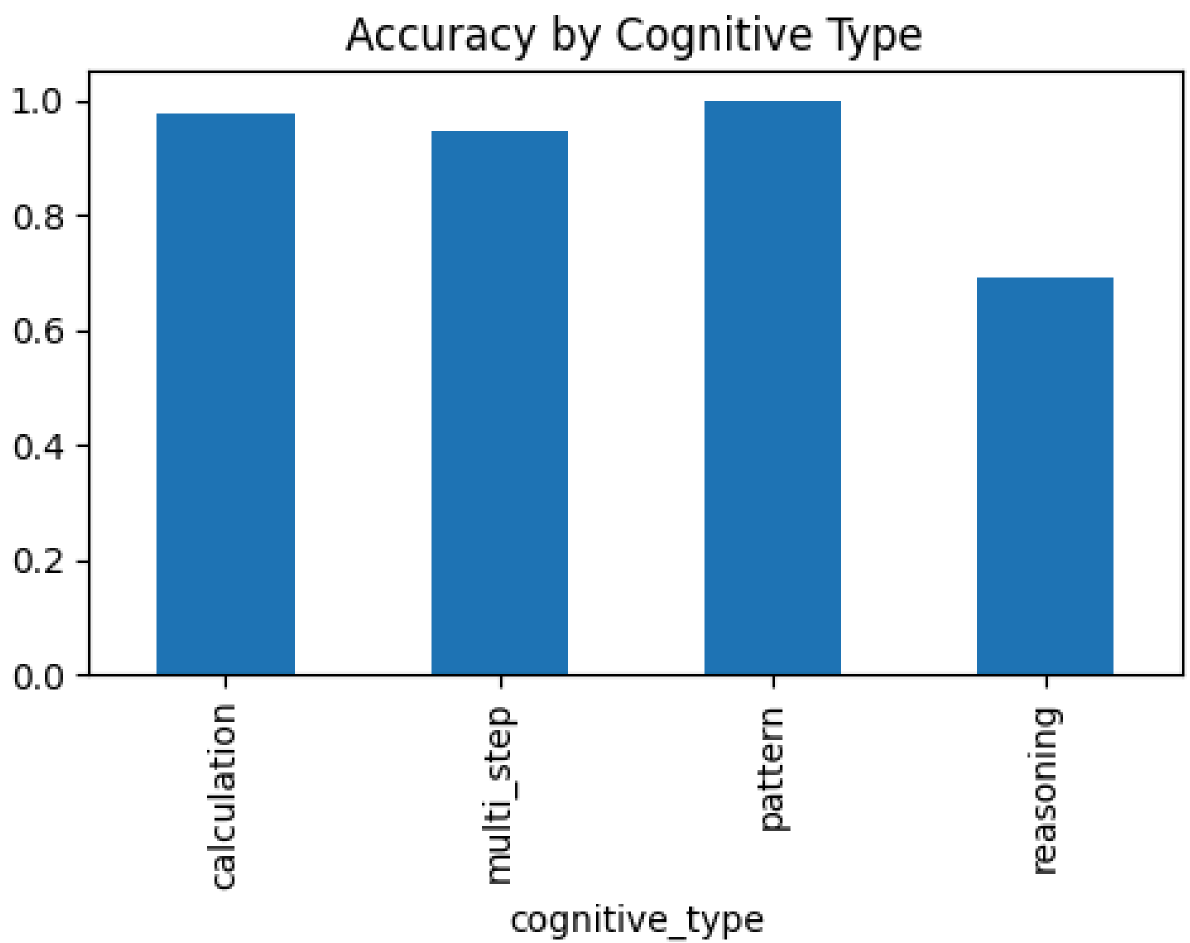
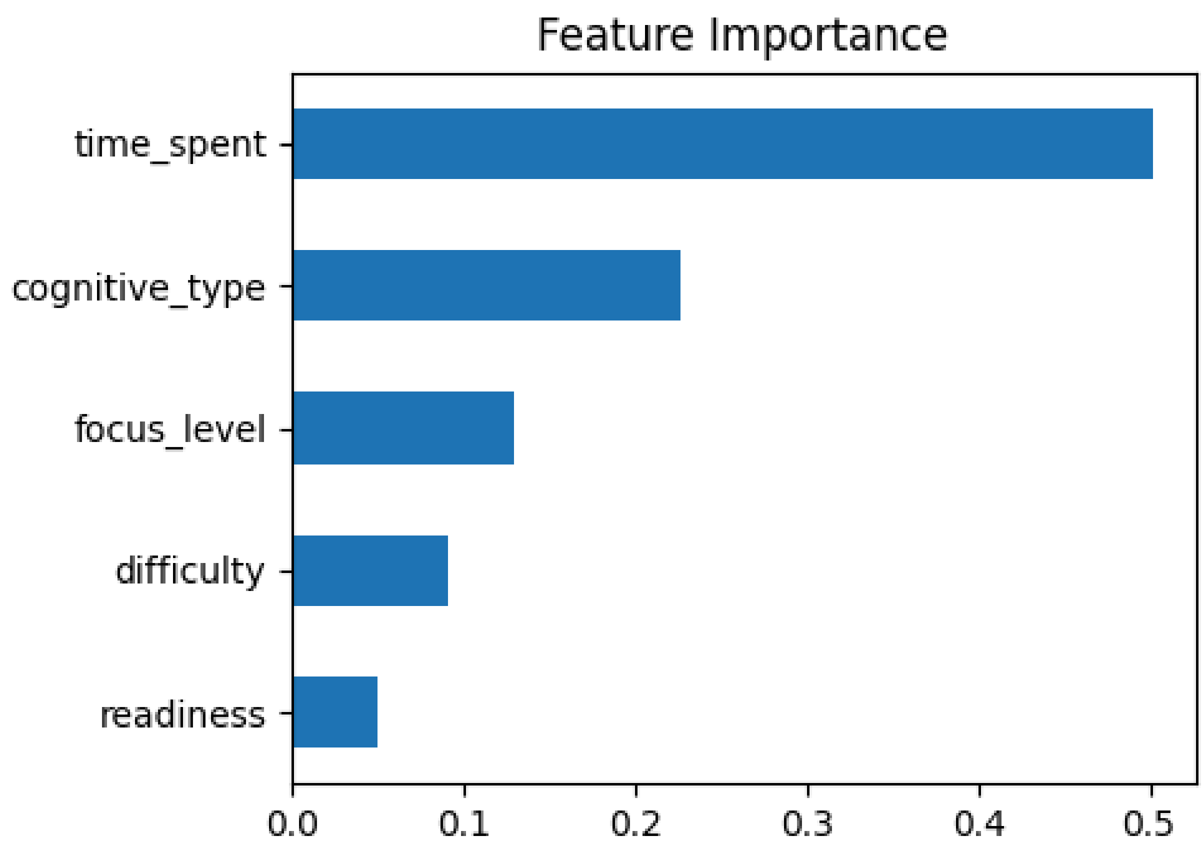


Figure 2: Statistical Feature Importance (Random Forest Weight Distribution)



TECHNICAL IMPLEMENTATION SUMMARY

Data acquisition was executed using a responsive frontend built on standard interactive layouts communicating with a custom Python-Flask backend and a relational MySQL data layer. Pipeline analysis utilized the standard Python data science stack (Pandas, NumPy, Scikit-Learn), transforming categorical variables via numerical Label Encoding before initializing a Random Forest classification script to process task outcomes based on behavioral state records.

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Edge AI Object Classifier

Win Research Centre (WRC) | ML Internship Suite

Architected and deployed a high-accuracy, resource-optimized image classification framework for the **Fit Choice World (FCW)** fitness ecosystem. The core engineering focus centered on balancing deep learning capabilities with the strict computing limitations of mobile hardware devices to automate user health logging.

MODEL FOOTPRINT

3.7 MB

INFERENCE LATENCY

~30 ms

FINALIZED ACCURACY

97.0%

CORPORATE SCORE

100/100

OPTIMIZATION ENGINEERING SOLUTIONS

- **Model Architecture Core (Transfer Learning):** Leveraged a fine-tuned MobileNetV2 architecture via TensorFlow/Keras, selecting it specifically for its processing efficiency on memory-constrained edge hardware.

- **Post-Training Quantization Layer:** Successfully executed integer quantization scripts via TensorFlow Lite (TFLite), compressing binary parameters into a 3.7 MB footprint to allow real-time on-device inference.
- **Overfitting & Bias Mitigation:** Identified initial convergence instability within the 'Random' and 'Food' categories during Week 5. Mitigated this variance by scaling the training collection and introducing data augmentation routines (rotation, scaling, and horizontal flips).
- **Production Routing Optimization:** Debugged local Flask backend file-upload logic to parse diverse image buffers and accelerated the prediction pipeline to guarantee latency remained strictly below a ~30ms production threshold.

EMPIRICAL DIAGNOSTICS & VISUAL EVIDENCE

The model validation pipeline evaluated structural generalization parameters across balanced test sets (150 total media evaluations; 50 samples per targeted category) to verify precision boundaries.

Figure 1: Confusion Matrix and Validation Classification Report

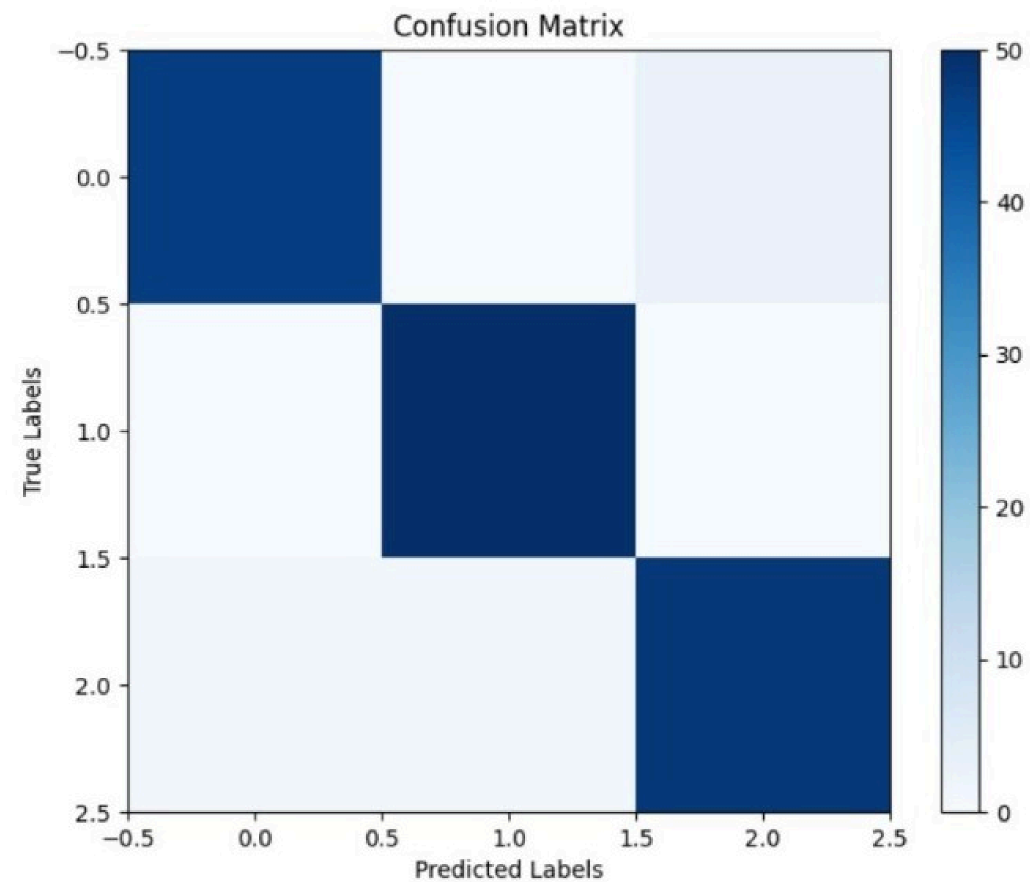

Fig 1: Mathematical matrix confirming a clean 0.97 macro average across food (1.00 recall) and miscellaneous classes.

Figure 2: Production Flask Web Server Interaction Log

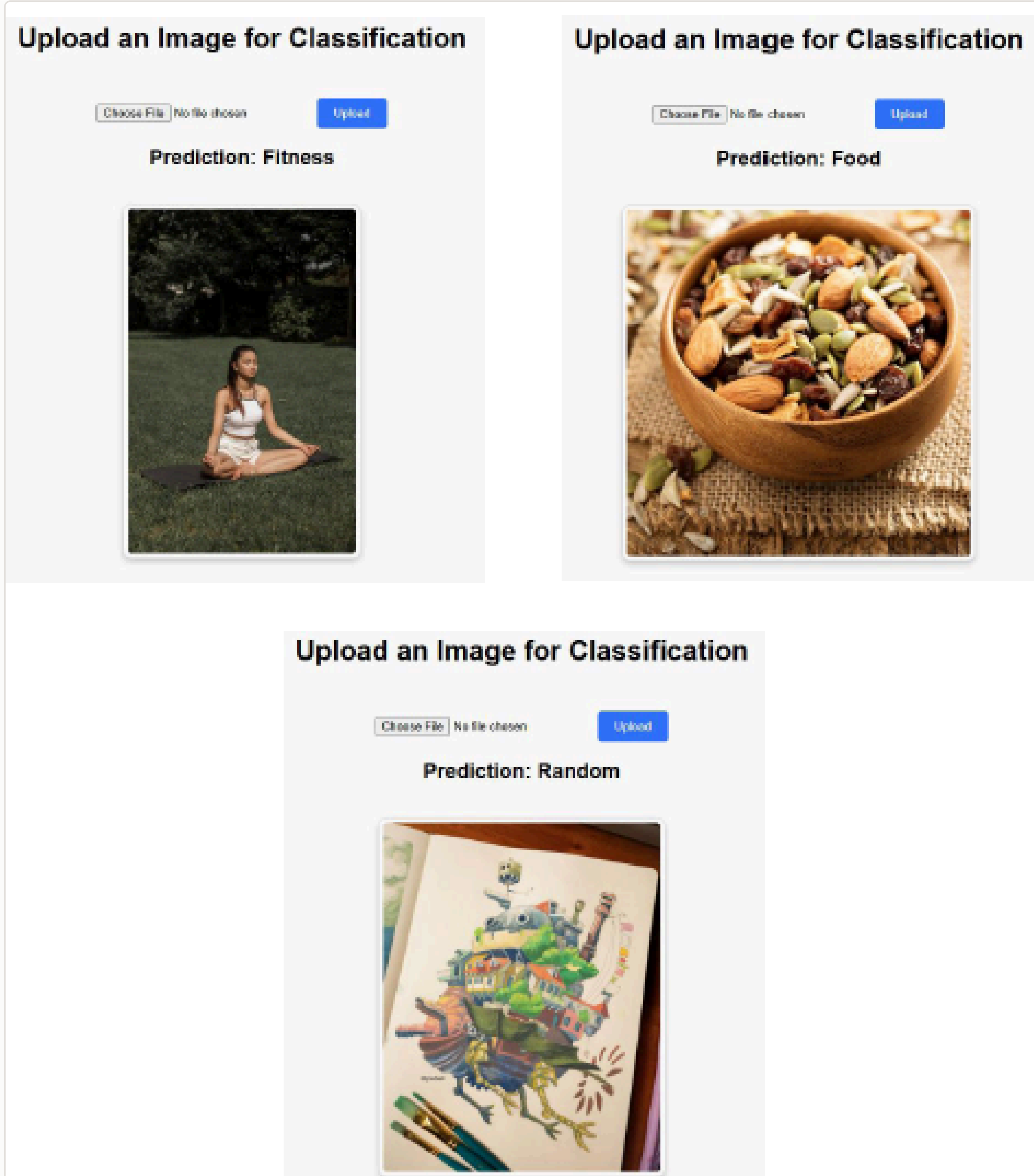
Upload an Image for Classification
Choose File No file chosen
Upload
Prediction: Fitness
Upload an Image for Classification
Choose File No file chosen
Upload
Prediction: Food
Upload an Image for Classification
Choose File No file chosen
Upload
Prediction: Random

Fig 2: Verified endpoint testing interface capturing instantaneous domain mapping across varied image streams.

TECHNICAL IMPLEMENTATION SUMMARY

The final pipeline utilizes an interactive, responsive frontend that routes image data straight to an optimized Python-Flask web server. The backend initializes the quantized model weights via the TensorFlow Lite Interpreter, processing image inputs and returning high-precision category tags instantly to enable automatic fitness tracking.

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Longitudinal Behavioral Analytics

Interdisciplinary Case Study | Phase 1 Synthesized

This multi-disciplinary project cross-analyzes a 10-column structured media consumption log against qualitative, coded personal journals spanning a consecutive 6-month timeline. Combining perspectives from Data Science, Psychology, Cognitive Science, and Human-Computer Interaction, the study maps behavioral trends and explores how individual media preferences coincide with changing emotional states and academic stressors over time.

TIMELINE SCOPE

6 Months

OBSERVATION GROUP

Single-Subject (N=1)

PEAK QUANTITY VOLUME

20 Titles (May)

BASELINE QUANTITY VOLUME

5 Titles (March)

INVESTIGATED RESEARCH QUESTIONS (RQS)

The research matrix sets up 7 distinct exploratory objectives to systematically decode cross-dataset associations:

- **Descriptive Parameters (RQ1–RQ2):** Quantified month-to-month volumetric fluctuations in media usage alongside evolving cognitive focus tracks.
- **Structural Association (RQ3–RQ4):** Evaluated non-random coincidences between regional platforms and persistent psychological constructs.
- **Media Psychology Frameworks (RQ5–RQ6):** Tested whether consumption themes mirrored or contrasted with dominant real-life stressors, evaluating the functional coping role of entertainment.
- **Longitudinal Synthesis (RQ7):** Tracked the co-variation of media metrics and emotional profiles across the entire 26-week horizon.

VALIDATED RESEARCH INSIGHTS

- **The Satiation Deficit (H12 - Supported):** Higher quantity does not equal greater satisfaction. May recorded peak velocity (20 titles) but hit the lowest satisfaction rating (3.62/5.0) and highest DNF rate. Conversely, March achieved peak satisfaction (4.20/5.0) despite recording the lowest volume (5 titles).
- **The Emotional Contrast Filter (H4 - Supported):** Media selections frequently contrasted with ongoing negative experiences. During months characterized by acute academic deadlines and physical fatigue, relationship-focused, healing, and romantic Asian narratives functioned as an active cognitive buffer.
- **The Invariant Factor (H10 - Supported):** Isolated *Hope* as the single dominant, stable emotional construct remaining consistently present across all six observed months regardless of baseline stress fluctuations.
- **Gradual Identity Development:** Personal reflections demonstrated a clear longitudinal movement, shifting from general future planning toward a stronger research identity and academic independence.

METHODOLOGICAL BOUNDARIES & LIMITATIONS

As a transparent, rigorous scientific pipeline, the study explicitly identifies the following boundary parameters:

- **Single-Participant Baseline:** The records reflect an individual case study; observations represent individual behavioral dynamics and cannot be generalized to population-level behavior.
- **Temporal Scale Constraints:** The capture layer is bounded at 6 consecutive months, creating an exploratory overview rather than a multi-year seasonal assessment.
- **Subjective Self-Report Boundaries:** Both consumption categories and qualitative psychological experiences rely on self-reported logs subject to personal internal assessment interpretation constraints.

PHASE 2 ACTIVE ENGINEERING EXTENSIONS

To address the data boundaries of manual exploration, current active development is deployed across two technical machine learning pipelines:

- **Automated Natural Language Processing (NLP):** Coding localized Python scripts deploying text-processing algorithms to replace manual content indexing, automating emotional keyword parsing and sentiment extraction straight from raw journal rows.
- **Predictive Data Science Modeling:** Designing rolling time-series classification models to investigate if media selection trends can anticipate changes in subjective user satisfaction vectors over time.